FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $t = 20 \text{ min} = \frac{1}{3}\text{hr}$

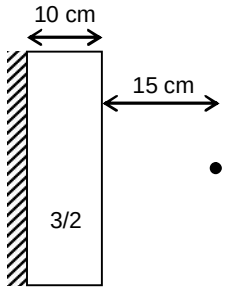
$1 \text{ km} = \sqrt{(6)^2 - V^2} \times \left(\frac{1}{3}\text{hr}\right)$

$V = 3\sqrt{3} \text{ km/hr}$

2. C

Sol. $X_{app} = \frac{10}{3/2} + \frac{10}{3/2} + \frac{15}{1}$

$= \frac{40}{3} + 15 = 28.33$



3. C

Sol. For no deviation

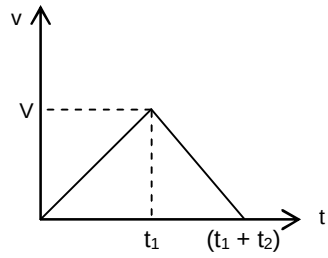
$(\mu_1 - 1)A_1 = (\mu_2 - 1)A_2$

$4^o(1.54 - 1) = (1.72 - 1)A_2$

$A_2 = \frac{4 \times 0.54}{0.72} = 3°.$

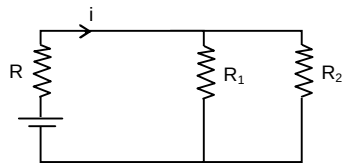
4. A

Sol. Required ratio = $\frac{V}{\text{Area}/(t_1+t_2)} = \frac{2}{1}$



5. C

Sol. Inductor behaves like a cell of emf, ε' = Blv and equivalent ckt is



Blv = ε'

Hence, $i = \frac{Blv}{R + \frac{R_1R_2}{R_1+R_2}}$

$= \frac{Blv(R_1+R_2)}{RR_1+RR_2+R_1R_2}$

6. C

Sol. Variation of moment of inertia with temperature

$I = I_o(1 + 2\alpha\,\Delta T)$

C.O.A.M:

$I_i\omega_i = I_f\,\omega_f$

7. D

Sol. Upon earthing, final potential of conductor becomes zero. Final charge may or may not be zero.

8. B

Sol. $\tan\theta = \frac{X_c - X_L}{X_R}$ and $\cos\theta$ = power factor.

9. A

Sol. $r = \frac{mv}{qB} = \frac{1\times 10}{1\times 2} = 5m$ And motion of particle will be in x – y plane.

10. B

Sol. $\frac{q}{4\pi\in_0 r} + \frac{1}{4\pi\varepsilon_o}\frac{(2a\,q)\cos\theta}{r^2}$ s

11. D

Sol. $W_{agent} - mgH = 0$

$W_{agent} = mgH$

12. D

Sol. $A_1V_1 = A_2V_2 \Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right) \Rightarrow \frac{a}{A}\int_0^t dt = -A\int_{h_1}^{h_2}\frac{dh}{\sqrt{2gh}}$

13. B

Sol. If we consider a point very close to one wire than magnetic field due to this wire will dominate.

14. D

Sol. $\Delta q = \Delta u + \Delta w$

$\Rightarrow \quad \Delta U = -996\ J = \frac{nR}{\gamma - 1}\Delta T$

$\gamma = \frac{5}{3}$ for mono-atomic gas.

15. C

Sol. $i = \frac{E}{R}\left(1 - e^{-\frac{tR}{L}}\right)$

16. B

Sol. $Kx = 2 \times g = 20, \quad x = \frac{1}{2}m$

$5\ gx - \frac{1}{2}Kx^2 = \frac{1}{2}5V^2$

$V = 2\sqrt{2}\ m/s$

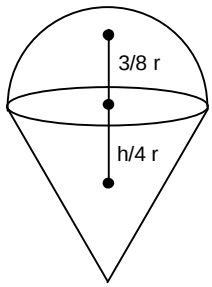
17. B

Sol. $F = \frac{dp}{dt} = \left(\sqrt{2}\,V\right)\frac{dm}{dt} = \left(\sqrt{2}\,V\rho\right)\frac{d(volume)}{dt} = \left(\sqrt{2}\,V\rho\right)(AV)$

18. B

Sol. $\frac{3}{8}r \cdot \rho\frac{2}{3}\pi r^3 = \frac{h}{4} \cdot \rho\frac{1}{3}\pi r^2 h$

$\Rightarrow \quad h = \sqrt{3}\,r$



19. A

Sol. $f\lambda = v = \frac{10\lambda v}{d} \Rightarrow \lambda = \frac{d}{10}.$

Maximum path diff. can be $\frac{d}{3}$. Hence, number of maximum will be 6.

20. B

Sol. According to Bohr's atomic model.

21. C

Sol. V = 0.6 m

$\Rightarrow \quad \frac{h_I}{h_o} = \frac{h_I}{0.05} = 2 \Rightarrow h_I = 0.1$ cm.

$\Rightarrow$ Distance between them = 0.3 cm = 3 mm.

22. B

Sol. $V_Q - V_O = \frac{1}{2}B(\omega)(r^2) = \frac{B\omega}{2}(2a)^2 = 2B\omega a^2$.

23. B

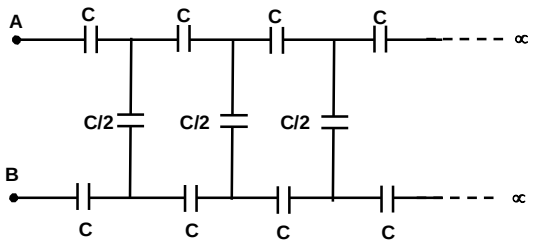
Sol. Total momentum = $\int_{\theta=-90}^{+90} (dm)v\cos\theta$

$dm = \frac{M}{\pi R} \times Rd\theta$

Total momentum = $\int \frac{M}{\pi R} vRd\theta \cos\theta = \frac{2Mv}{\pi}$

24. C

Sol. The Equivalent circuit is as follows



25. A

Sol. $I_a = \frac{3V}{15R}, I_b = \frac{3V}{5R}$

$I_c = \frac{6V}{15R}$.

26. C

Sol. At first the pressure will increase rapidly and then it will become constant.

27. B

Sol. $dV = -IdR = = -I\rho\frac{dx}{A}$

$\frac{dV}{dx} = -I\frac{\rho}{A} = -k$

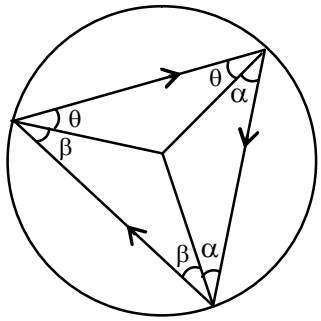
$\Rightarrow R = k\ell/I$

II Method:

$$V_2 - V_1 = \int \left(\frac{dV}{d\ell}\right) d\ell = kL = IR$$

28. D

Sol. $\tan\alpha = \dfrac{u\sin\theta}{eu\cos\theta} = \dfrac{\tan\theta}{e}$



$\therefore \tan\alpha = \dfrac{1}{e}\tan\theta$

$\therefore \tan\beta = \dfrac{1}{e}\tan\theta$

$2\theta + 2\alpha + 2\beta = 180°$

$\alpha + \beta = (90 - \theta)$

$\tan(\alpha + \beta) = \cot\theta$

$$\Rightarrow \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{1}{\tan\theta}$$

$$\Rightarrow \frac{\frac{1}{e}\tan\theta + \frac{1}{e^2}\tan\theta}{1 - \frac{1}{e^3}\tan^2\theta} = \frac{1}{\tan\theta}$$

$$\Rightarrow \tan^2\theta\left(\frac{1}{e} + \frac{1}{e^2}\right) = 1 - \frac{1}{e^3}\tan^2\theta$$

$$\Rightarrow \tan^2\theta\left(\frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3}\right) = 1$$

$$\tan^2\theta\left(\frac{1}{\frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3}}\right) = \frac{e^3}{1 + e + e^2}$$

29. C

Sol. $\because$ acceleration = 0. Net force towards left = Net force towards right.

30. A

Sol. $\Delta m = 0.1$ kg ; $\dfrac{\Delta m}{m} = \dfrac{0.1}{10}$

$\therefore$ Error = $\pm\dfrac{\Delta m}{m} \times 100 = \pm 1\%$.

Chemistry

PART – II

SECTION – A

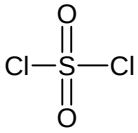
31. A

Sol. Stability of anion $= ClO_4^- > ClO_3^- > ClO_2^- > ClO^-$

Oxidation number of Cl $= \overset{+7}{H}ClO_4 > \overset{+5}{H}ClO_3 > \overset{+3}{H}ClO_2 > \overset{+1}{H}OCl$

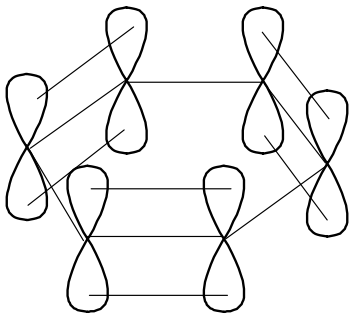
32. D

Sol.



33. A

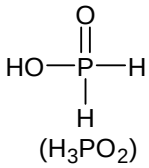
Sol.



Molecular plane is the nodal plane of all the π-bonds.

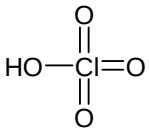
34. D

Sol.



(H_3PO_2)

monobasic nature and replaceable hydrogen is one



monobasic nature and replaceable hydrogen is one

35. D

Sol. $\left[Co(NH_3)_4(H_2O)_2\right]Cl_2$ has two geometrical isomers and Co^{2+} has $3d^7$ configuration (paramagnetic).

36. D

Sol. (A) $\left[Co(NH_3)_6\right]Cl_3$

(B) $\left[Co(NH_3)_4Cl_2\right]Cl$

(C) $\left[Co(NH_3)_5Cl\right]Cl_2$

(D) $\left[Co(NH_3)_3Cl_3\right]$

37. A

Sol. • Ag_2S, Cu_2S, PbS are the main sulphide ore of Ag, Cu and Pb respectively

- Sn exists as SnO_2
- Al exists as Al_2O_3
- Mg exists as $MgSO_4 \cdot 7H_2O$, $MgCO_3$, etc.

38. A

Sol. Only five HCO_3^- salt exist in solid form $NaHCO_3$, $KHCO_3$, $RbHCO_3$, $CsHCO_3$, NH_4HCO_3.

39. C

Sol. $Ca_3(PO_4)_2$ does not give volatile product

$$CO_3^{2-} \xrightarrow{HNO_3} CO_2$$

$$NO_2^- \xrightarrow{HNO_3} NO \uparrow \xrightarrow{O_2} NO_2 \uparrow$$

$$Br^- \xrightarrow{HNO_3} Br_2 \uparrow$$

40. D

Sol. Hard water contains soluble salt of Ca^{2+} and Mg^{2+}.
$CaCO_3$ = Insoluble salt.

41. D

Sol. Total moles of H_2SO_4 = 0.1 mole and total mass taken 100 × 1.5 = 150 g

$$\text{Final volume} = \frac{150+400}{1.25} = \frac{550}{1.25} = 440$$

$$\therefore \quad M = \frac{0.1}{440} \times 1000 = \frac{1}{4.4} = 0.227 \text{ M}$$

42. A

Sol. $\overset{+6}{Cr_2}O_7^{2-} \longrightarrow \overset{+6}{Cr}O_4^{2-}$

43. C

Sol. When an electron of charge e and mass m is accelerated with a potential difference of V volts K.E. = eV

$$\Rightarrow \quad \frac{1}{2}mv^2 = eV \quad \text{or} \quad v^2 = \frac{2\ eV}{m}$$

$$\Rightarrow v = \sqrt{\frac{2\ eV}{m}}$$

44. D

Sol. $$E = \frac{1240}{\lambda_{nm}} = \frac{1240}{91} = 13.6 \text{ eV}$$

45. A

Sol. Moles of C_2H_6 and C_2H_4 are a and b respectively.

$$a + b = \frac{28}{22.4} = 1.25$$

$$\underset{a}{C_2H_6}(g) + 3.5\ \underset{3.5a}{O_2}(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$$

$$\underset{b}{C_2H_4}(g) + 3\ \underset{3b}{O_2}(g) \longrightarrow 2CO_2(g) + 2H_2O(\ell)$$

Moles of O_2 = 3.5 a + 3b = 4

a + b = 1.25

a = 0.5

b = 0.75

46. B

Sol. w = – Area of P – V curve

47. A

Sol. From graph, $[A]_{eq}$ = 0.1 M, $[B]_{eq}$ = 0.4 M

$$K_c = \frac{[B]^2}{[A]^1} = 1.6$$

48. C

Sol. $\underset{C-x}{HCOOH} \rightleftharpoons \underset{x+0.01}{H^+} + \underset{x}{HCOO^-}$

$$K_a = \frac{(x + 0.01)x}{0.01 - x}$$

neglect x w.r.t. 0.01 due to common ion effect.

$K_a = x$

$$\therefore x \simeq [HCOO^-] = 1.8 \times 10^{-4}$$

49. C

Sol. $$\text{Rate of reaction} = \frac{\text{Rate of disappearance / appearance}}{\text{Stoichiometric coefficient}}$$

50. C

Sol. $Cu(s)\,|\,Cu^{2+}(aq)\,||\,Cl^-(aq)\,|\,Hg_2Cl_2(s)\,|\,Hg(\ell)\,|\,Pt(s)$,

Anode : $Cu\,|\,Cu^{2+}$

Cathode : $Cl^-\,|\,Hg_2Cl_2\,|\,Hg$

51. A

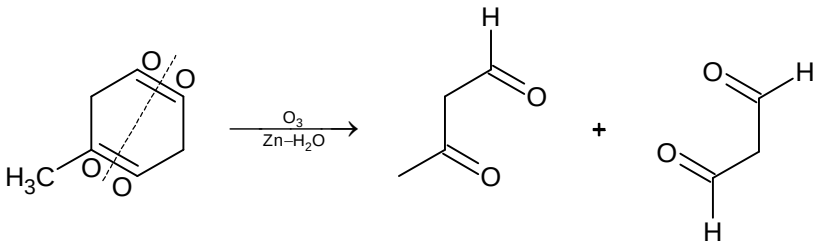
Sol. $i = 1 + (n-1)\alpha$ so $n = \frac{i + \alpha - 1}{\alpha}$

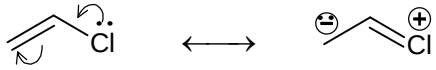
52. A

Sol. The co-ordination number of a metal crystallising in a hexagonal close-packed structure is 12.

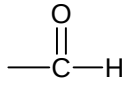
53. D
Sol. All the above options are correct.

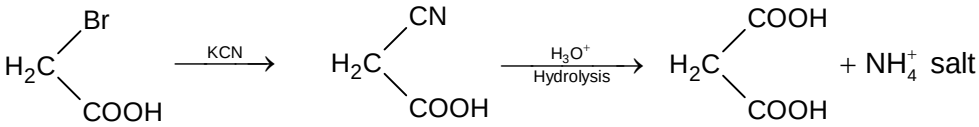
54. B
Sol. $CH_3 - CH = CH - CH_3 \xrightarrow[H^+]{KMnO_4} CH_3COOH + HOOC - CH_3$

55. A
Sol. 

56. D
Sol. 

57. B
Sol. SeO_2 is a selective oxidising agent.

58. D
Sol. It is given by  group

59. B
Sol. 

60. A
Sol. $PhCOOH + NaHCO_3 \longrightarrow PhCOONa(\text{Solution}) + CO_2 \uparrow + H_2O$

Mathematics

PART – III

SECTION – A

61. A

Sol. $\because 9^x - 3^x + 1 = \left(3^x - \frac{1}{2}\right)^2 - \frac{1}{4} + 1 = \left(3^x - \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$

$\therefore$ For $x \in (-\infty, 1)$

$\Rightarrow \quad \frac{3}{4} \leq 9^x - 3^x + 1 < \frac{28}{4}$

Now, $f(x) = [9^x - 3^x + 1] = \{0, 1, 2, 3, 4, 5, 6\}$

$\therefore$ $R_f = \{0, 1, 2, 3, 4, 5, 6\}$.

62. C

Sol. Sum of roots $= |\alpha| + |\beta| = -\alpha - \beta (\because \alpha < 0 \text{ and } \beta < 0)$

$= -(\alpha + \beta) = -\left(-\frac{b}{a}\right) = \frac{b}{a}$

$= \left|\frac{b}{a}\right| (\because |\alpha| + |\beta| > 0)$

and product of roots $= |\alpha||\beta| = |\alpha\beta| = \left|\frac{c}{a}\right|$

Hence, required equation is

$x^2 - \left|\frac{b}{a}\right| x + \left|\frac{c}{a}\right| = 0$ or $|a|x^2 - |b|x + |c| = 0$

63. A

Sol. Let $P = \sum_{i=0}^{n-1} \frac{\alpha_i}{(3 - \alpha_i)} = -\sum_{i=0}^{n-1} \frac{(3 - \alpha_i) - 3}{(3 - \alpha_i)}$

$= 3\sum_{i=0}^{n-1} \frac{1}{(3 - \alpha_i)} - \sum_{i=0}^{n-1} 1$ … (i)

$\because z^n - 1 = \prod_{i=0}^{n-1} (z - \alpha_i)$

$\because \ln (z^n - 1) = \sum_{i=0}^{n-1} \ln (z - \alpha_i)$

Differentiating both sides w.r.to z, we get

$\frac{nz^{n-1}}{(z^n - 1)} = \sum_{i=0}^{n-1} \frac{1}{(z - \alpha_i)}$ … (ii)

and put z = 3, then

$\frac{n \cdot 3^{n-1}}{(3^n - 1)} = \sum_{i=0}^{n-1} \frac{1}{(3 - \alpha_i)}$

From eq. (i), $3\left(\frac{n \cdot 3^{n-1}}{3^n - 1}\right) - n = \frac{n \cdot 3^n}{3^n - 1} - n = \frac{n}{(3^n - 1)}$.

64. C

Sol. Let $A = m\lambda$, $H = n\lambda$

$\therefore G^2 = AH = mn\lambda^2$

$\because$ a, b are the roots of $t^2 - (a + b)t + ab = 0$

or $t^2 - 2m\lambda t + mn\lambda^2 = 0$

$$\therefore t = \frac{2m\lambda \pm \sqrt{(4m^2\lambda^2 - 4mn\lambda^2)}}{2}$$

$$= \lambda\sqrt{m}\{\sqrt{m} \pm \sqrt{(m-n)}\}$$

$$a : b = \sqrt{m} + \sqrt{(m-n)} : \sqrt{m} - \sqrt{(m-n)}$$

or $\sqrt{m} - \sqrt{m-n} : \sqrt{m} + \sqrt{m-n}$

65. C

Sol. $\because$ $$\tan y = \frac{1+\sqrt{(1-x)}}{1+\sqrt{(1+x)}}$$

In R.H.S. put $x = \cos\theta$, then

$$\tan y = \frac{1+\sqrt{2}\sin(\theta/2)}{1+\sqrt{2}\cos(\theta/2)} = \frac{\frac{1}{\sqrt{2}} + \sin(\theta/2)}{\frac{1}{\sqrt{2}} + \cos(\theta/2)}$$

$$= \frac{\sin\left(\frac{\pi}{4}\right) + \sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\pi}{4}\right) + \cos\left(\frac{\theta}{2}\right)} = \frac{2\sin\left(\frac{\pi}{8} + \frac{\theta}{4}\right)\cos\left(\frac{\pi}{8} - \frac{\theta}{4}\right)}{2\cos\left(\frac{\pi}{8} + \frac{\theta}{4}\right)\cos\left(\frac{\pi}{8} - \frac{\theta}{4}\right)} = \tan\left(\frac{\pi}{8} + \frac{\theta}{4}\right)$$

$$\therefore \quad y = n\pi + \frac{\pi}{8} + \frac{\theta}{4}$$

$$\Rightarrow \quad 4y = 4n\pi + \frac{\pi}{2} + \theta$$

$\therefore \quad \sin 4y = \cos\theta = x$

$\Rightarrow \quad \sin x = 1$ (impossible)

Case III : $\sin x = 1$

$\therefore \quad y = 1$

i.e., $\cos x = 1$

$\Rightarrow \quad \sin x = 0$ (impossible)

Hence, no solution

i.e. no. of solutions is zero.

66. A

Sol. $\because$ $$y = \frac{1}{3} = [\sin x + [\sin x + [\sin x]]]$$

$$= \frac{1}{3}([\sin x] + [\sin x] + [\sin x])$$

$$= [\sin x]$$

and $[y + [y]] = 2\cos x$

$\Rightarrow \quad 2[y] = (2\cos x)$

$\Rightarrow \quad [y] = \cos x$

Case I : When $-1 \le \sin x < 0$

$\Rightarrow \quad [\sin x] = -1$

Then, $y = -1$

i.e., $[-1] = \cos x$

$\Rightarrow \quad \cos x = -1$

$\Rightarrow \quad \sin x = 0$ (impossible)

Case II : When $0 \le \sin x < 1$

$\therefore \quad [\sin x] = 0$

then, $y = 0$

i.e., $0 = \cos x$

67. A

Sol. $\therefore f(x+y) = f(x) \cdot f(y)$

for $x = y = 1$

$f(2) = 2^2$

for $x = 1, y = 2$

$f(3) = f(1)f(2) = 2^3$

$\therefore \quad f(n) = 2^n$

$\therefore \quad a = f(3) = 2^3 = 8,$

$b = f(1) + f(3) = 2 + 8 = 10$

and $c = f(2) + f(3) = 2^2 + 2^3 = 12$

$\cos A = \frac{3}{4}$ and $\cos C = \frac{1}{8}$

$\cos 2A = 2\cos^2 A - 1 = 2 \times \frac{9}{16} - 1 = \frac{1}{8} = \cos C$

$\therefore 2A = C$

68. A

Sol. $y = \cos x$

$y = \sin 3x$

$\therefore \sin 3x = \cos x = \sin\left(\frac{\pi}{2} - x\right)$

$\Rightarrow \quad 3x = n\pi + (-1)^n\left(\frac{\pi}{2} - x\right)$

Case I: when n is even, then

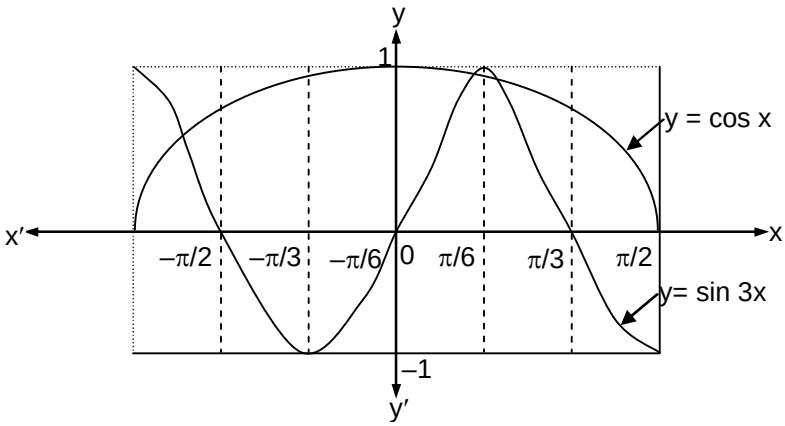
$3x = n\pi + \frac{\pi}{2} - x$

$\Rightarrow \quad x = \frac{n\pi}{4} + \frac{\pi}{8}$

$\therefore \quad x = \frac{\pi}{8}$, at $n = 0$

Then, $y = \cos\frac{\pi}{8}$

$\Rightarrow \left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$



Case II : when n is odd, then

$3x = n\pi - \frac{\pi}{2} + x$

or $x = \frac{n\pi}{2} - \frac{\pi}{4}$

$= \frac{\pi}{4} (\text{for } n = 1)$

Then, $y = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\Rightarrow \left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

Hence, point of intersection is

$\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ and $\left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$

69. D

Sol. Required ways $= ({}^3C_1 \times {}^2C_1 \times {}^1C_1) \times 2 = 12$

70. D

Sol. $(1+\lambda)^n (\mu+1)^n (\lambda+\mu)^n = \sum_{r=0}^{n} C_r \lambda^r \sum_{s=0}^{n} C_s \mu^{n-s} \sum_{t=0}^{n} C_t \lambda^{n-1} \mu^t$

Since, C_0^3 is the coefficient of $\lambda^0 \mu^{n-0} \lambda^{n-0} \mu^0$

i.e., $\lambda^n \mu^n (r = s = t = 0)$

Now, C_1^3 is the coefficient of $\lambda^1 \mu^{n-1} \lambda^{n-1} \mu^1$

i.e., $\lambda^n \mu^n (r = s = t = 1)$ etc.

and C_k^3 is the coefficient of $\lambda^k \mu^{n-k} \lambda^{n-k} \mu^k$

i.e., $\lambda^n \mu^n (r = s = t = k)$

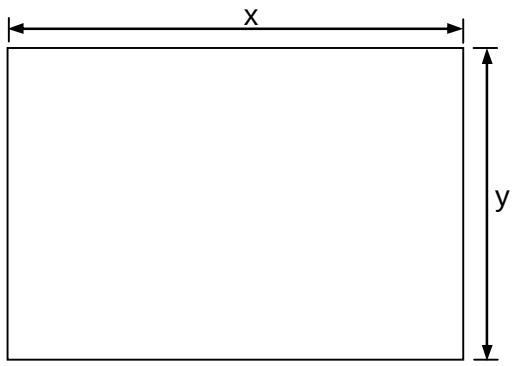
Hence, the coefficient of $\lambda^n\mu^n$ in

$$(1+\lambda)^n(\mu+1)^n(\lambda+\mu)^n = C_0^3 + C_1^3 + C_2^3 + ... + C_n^3 = \sum_{r=0}^{n} C_r^3$$

71. C

Sol. $\because 2x + 2y = 4L$

$\Rightarrow x + y = 2L$



$\therefore$ The are of the rectangle = xy

$= x(2L - x)$

But given, $x(2L - x) \le \frac{L^2}{4}$

$$\Rightarrow \quad x^2 - 2Lx + \frac{L^2}{4} \ge 0$$

$$\Rightarrow \quad (x-L)^2 - L^2 + \frac{L^2}{4} \ge 0$$

$$\Rightarrow \quad (x-L)^2 - \left(\frac{L\sqrt{3}}{2}\right)^2 \ge 0$$

$$\left(x - L + \frac{L\sqrt{3}}{2}\right)\left(x - L - \frac{L\sqrt{3}}{2}\right) \ge 0$$

$$\left\{x - \left(\frac{2-\sqrt{3}}{2}\right)L\right\}\left\{x - \left(\frac{2+\sqrt{3}}{2}\right)L\right\} \ge 0$$

$$\therefore x \in \left[0\left(,\frac{2-\sqrt{3}}{2}\right)L\right] \cup \left[\left(\frac{2+\sqrt{3}}{2}\right)L,\ 2L\right]$$

Now, x takes values in the interval [0, 2L]

$\therefore$ Required probability

$$= \frac{\int_0^{\left(\frac{2-\sqrt{3}}{2}\right)L} dx + \int_{\left(\frac{2+\sqrt{3}}{2}\right)L}^{2L} dx}{\int_0^{2L} dx} = \frac{\left(\frac{2-\sqrt{3}}{2}\right)L + 2L - \left(\frac{2+\sqrt{3}}{2}\right)L}{2L - 0} = \frac{2L - \sqrt{3}L}{2L} = \left(\frac{2-\sqrt{3}}{2}\right)$$

72. C

Sol. Possible values are –2, –1, 0, 1, 2

i.e., $\begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1, \begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} = 0, \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$

$\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2, \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2$

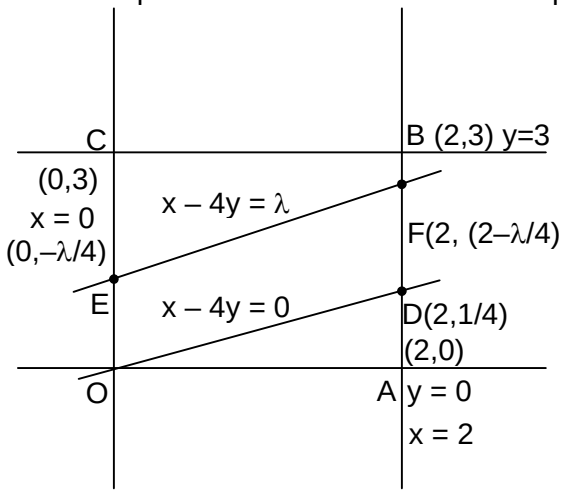
73. D

Sol. $\det(AA^{-1}) = \det(I)$

$\Rightarrow \det(A)\det(A^{-1}) = 1 \Rightarrow \det(A^{-1}) = \dfrac{1}{\det(A)}$

74. B

Sol. $\therefore$ Area of quadrilateral OABC = 2×3 =6 sq. units



$\therefore$ Area of OAFEO = $\dfrac{1}{2}$ Area of quadrilateral OABC

$\Rightarrow \quad \dfrac{1}{2} \times \left(\dfrac{2-\lambda}{4} - \dfrac{\lambda}{4} \right) \times 2 = 3 \Rightarrow \dfrac{1}{2} - \dfrac{\lambda}{2} = 3$

or $\quad -\dfrac{\lambda}{2} = \dfrac{5}{2}$

$\therefore \quad = -5$

$\therefore$ Required equation is $x - 4y + 5 = 0$

75. C

Sol. $\because \quad y = |x-1|$

$\Rightarrow \quad y^2 = (x-1)^2$

Change x by –x, then required image is

$y^2 = (-x-1)^2$

$\Rightarrow \quad y^2 = x^2 + 1 + 2x$

or $\quad x^2 - y^2 + 2x + 1 = 0$

76. D

Sol. The image of the circle has same radius but centre different. If centre is (α, β), then

$$\frac{\alpha-3}{1}=\frac{\beta-2}{1}=\frac{-2(3+2-19)}{1^2+1^2}$$

$\Rightarrow \quad \alpha-3=\beta-2=14$

$\therefore \quad \alpha=17$

$\beta=16$

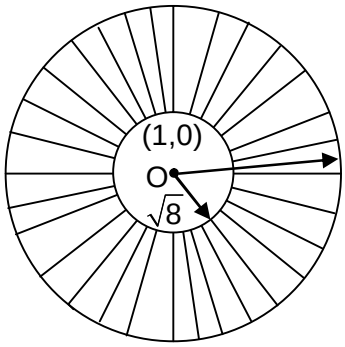
$\therefore$ Required circle is $(x-17)^2+(y-16)^2=1$

77. D

Sol. $\because$ The point ([P + 1], [P]) lies inside the circle

$x^2+y^2-2x-15=0$, then

$[P+1]^2+[P]^2-2[P+1]-15<0$



$\Rightarrow \quad ([P]+1)^2+[P]^2-2([P]+1)-15<0$

$\Rightarrow \quad 2[P]^2-16<0$

$\Rightarrow \quad [P]^2<8$ … (i)

$\because$ Circles are concentric

$\therefore$ Point ([P + 1], [P]) lies out side the circle

$x^2+y^2-2x-7=0$

$\therefore \quad ([P+1])^2+[P]^2-2[P+1]-7>0$

$\Rightarrow \quad ([P+1])^2+[P]^2-2[P]-2-7)=0$

$[P]^2+2[P]+1+[P]^2-2[P]-9=0$

$\therefore \quad 2[P]^2-8>0$

$[P]^2>4$ … (ii)

From eqs. (i) and (ii), we get

$4<[P]^2<8$ which is impossible

$\therefore$ For no value of 'P', the point will be within the region.

78. B

Sol. Tangent of parabola $y^2=4ax$ is

$$y=mx+\frac{a}{m} \quad \text{… (i)}$$

Where $m=\tan\alpha$

And equation of normal parallel to eq. (i) is

$y=mx-2am-am^3$ … (ii)

Distance between eqs. (i) and (ii) is $\dfrac{\left|\dfrac{a}{m}+2am+am^3\right|}{\sqrt{(1+m^2)}}$

$$=\frac{a(1+m^2)^2}{m\sqrt{(1+m^2)}}=\frac{a(1+m^2)^{3/2}}{m}=\frac{a(1+\tan^2\alpha)^{3/2}}{\tan\alpha}=\frac{a\sec^3\alpha}{\tan\alpha}=a\sec^2\alpha\cdot\operatorname{cosec}\alpha$$

79. C

Sol. The tangent at the point of shortest distance from the line x + y = 7 parallel to the given line.

Any point on the given ellipse is $(\sqrt{6}\cos\theta,\sqrt{3}\sin\theta)$.

Equation of the tangent is

$$\frac{x\cos\theta}{\sqrt{6}}+\frac{y\sin\theta}{\sqrt{3}}=1$$

It is parallel to x + y = 7

$$\Rightarrow \quad \frac{\cos\theta}{\sqrt{6}}=\frac{\sin\theta}{\sqrt{3}} \Rightarrow \quad \frac{\cos\theta}{\sqrt{2}}=\frac{\sin\theta}{1}=\frac{1}{\sqrt{3}}$$

The required point is (2, 1).

80. B

Sol. Equation of normal at $P(a\sec\phi,b\tan\phi)$ is

Hence, co-ordinates of L and M are

$$L\equiv\left(\frac{a^2+b^2}{a}\cdot\sec\phi,0\right)$$

and $M=\left(0,\dfrac{a^2+b^2}{b}\tan\phi\right)$ respectively.

Let mid-point of ML is Q (h, k)

Then $h=\dfrac{(a^2+b^2)}{2a}\sec\phi$

$$\therefore \quad \sec\phi=\frac{2ah}{(a^2+b^2)} \qquad \ldots \text{(i)}$$

and $k=\dfrac{(a^2+b^2)}{2b}\tan\phi$

$$\therefore \quad \tan\phi=\frac{2bk}{(a^2+b^2)} \qquad \ldots \text{(ii)}$$

From eqns. (i) and (ii)

$$\sec^2\phi-\tan^2\phi=\frac{4a^2h^2}{(a^2+b^2)^2}-\frac{4b^2k^2}{(a^2+b^2)^2}$$

Hence, locus

$$\frac{x^2}{\left(\dfrac{a^2+b^2}{2a}\right)^2}-\frac{y^2}{\left(\dfrac{a^2+b^2}{2b}\right)^2}=1$$

Let eccentricity is e_1

$$\therefore \left(\frac{a^2+b^2}{2b}\right)=\left(\frac{a^2+b^2}{2a}\right)(e_1^2-1)$$

$$\Rightarrow \quad a^2=b^2(e_1^2-1)$$

$$\Rightarrow \quad a^2=a^2(e^2-1)(e_1^2-1)$$

$$(\because b^2=a^2(e^2-1))$$

$$\Rightarrow \quad e^2e_1^2-e^2-e_1^2+1=1$$

$$\Rightarrow \quad e_1^2(e^2-1)=e^2$$

$$\therefore \quad e_1=\frac{e}{\sqrt{(e^2-1)}}$$

81. A

Sol. $\because 0<\cos^2(n!\pi x)\le 1$

If $-0<\cos^2(n!\pi x)<1$

Then, $f(x)=\lim\limits_{m\to\infty}\lim\limits_{n\to\infty}\{1+(\cos^2(n!\pi x))^m\}$

= 1 + 0 = 1, when x is irrational

and if $\cos^2(n!\pi x)=1$

$$\Rightarrow \quad n!\pi x=r\pi$$

$$\Rightarrow x=\frac{r}{n!}=\text{rational}$$

Then, $f(x)=\lim\limits_{m\to\infty}\lim\limits_{n\to\infty}\{1+(\cos^2(n!\pi x))^m\}$ = 1 + 1 = 2

Hence, $f(x)=\begin{cases}2, x \text{ is rational}\\ 1, x \text{ is irrational}\end{cases}$

82. C

Sol. Given, $\sqrt{(x+y)}+\sqrt{(y-x)}=a$ ... (i)

$$\therefore \quad \sqrt{x+y}-\sqrt{y-x}=\frac{2x}{a} \qquad \text{... (ii)}$$

Adding eqns. (i) and (ii), then

$$2\sqrt{x+y}=a+\frac{2x}{a}$$

Squaring, $4x+4y=a^2+\frac{4x^2}{a^2}+4x$

$$\therefore \quad 4+4\frac{dy}{dx}=0+\frac{8x}{a^2}+4$$

$$\therefore \quad 0+4\frac{d^2y}{dx^2}=\frac{8}{a^2}$$

$$\therefore \quad \frac{d^2y}{dx^2}=\frac{2}{a^2}$$

83. A

Sol. $\because \quad 1\le|\sin x|+|\cos x|\le\sqrt{2}$

$\therefore \quad y = \left[|\sin x| + |\cos x|\right] = 1$

Let P and Q be the points of intersection of given curves.

Now, solving $y = 1$ and $x^2 + y^2 = 5$

$\therefore \quad x^2 + 1 = 5$

$\Rightarrow \quad x = \pm 2$

$\therefore \quad P \equiv (-2,1)$ and $Q \equiv (2,1)$

Clearly, the slope of line $y = 1$ is zero.

$\because \quad x^2 + y^2 = 5$

$\therefore \quad 2x + 2y\dfrac{dy}{dx} = 0 \Rightarrow \quad \dfrac{dy}{dx} = -\dfrac{x}{y}$

$\left(\dfrac{dy}{dx}\right)_{(-2,1)} = 2$ and

$\left(\dfrac{dy}{dx}\right)_{(2,1)} = -2$

84. B

Sol. $\int x^{-2/3} \cdot \{(x^{1/2})(x^{-1/2} + 1)\}^{-5/3} dx$

$= \int x^{-2/3} \cdot x^{-5/6} \cdot (x^{-1/2} + 1)^{-5/3} dx$

$= \int x^{-3/2} \cdot (x^{-1/2} + 1)^{-5/3} dx$

Put $x^{-1/2} + 1 = t$

$\therefore -\dfrac{1}{2} x^{-3/2} dx = dt$

Then, $-2\int t^{-5/3} dt = -2 \cdot \dfrac{t^{-2/3}}{-2/3} + C = 3(x^{-1/2} + 1)^{-2/3} + C$

85. A

Sol. $\left[\dfrac{2}{e^x}\right] = \begin{cases} 1, \text{ if } x \le \log_e 2 \\ 0, \text{ if } x > \log_e 2 \end{cases}$

$\therefore I = \int_0^{\log_e 2} \left[\dfrac{2}{e^x}\right] dx + \int_{\log_e 2}^{\infty} \left[\dfrac{2}{e^x}\right] dx$

$I = \int_0^{\log_e 2} 1 dx + 0 = \log_e 2$

86. A

Sol. Let $I = \displaystyle\int_2^3 \frac{dx}{\sqrt{(1+x^3)}}$

Since, $1 + x^3 > (1 + x)^2$ in $x \in (2, 3)$

$\Rightarrow \sqrt{(1+x^3)} > (1+x)$

$\int_2^3 \dfrac{1}{\sqrt{(1+x^3)}} dx < \int_2^3 \dfrac{1}{(1+x)} dx$

$\Rightarrow I < [\log_e(1+x)]_2^3 \Rightarrow I < \log_e(4/3)$

$\therefore I < 1$

87. A

Sol. We have, $\frac{dy}{dx} = 2x + 1$

$\Rightarrow y = x^2 + x + c$, it passes through (1, 2)

$\therefore c = 0$

Then , $y = x^2 + x$

$\therefore$ Required area $= \int_0^1 (x^2 + x)dx = \frac{5}{6}$ sq. units

88. A

Sol. The reciprocal of polar subtangent is $\frac{1}{r^2}\frac{dr}{d\theta}$.

According to the given condition,

$\frac{1}{r} + \frac{1}{r^2}\frac{dr}{d\theta} = k \Rightarrow d\theta = \frac{dr}{r(kr-1)} \Rightarrow kr - 1 = cre^{\theta}$

89. B

Sol. If DC's of line of intersection of planes4y + 6z = 5 and 2x + 3y + 5z = 5 are l, m, n

$\therefore \quad 0 + 4m + 6n = 0$

$2l + 3m + 5n = 0$

$\Rightarrow \quad \frac{l}{1} = \frac{m}{6} = \frac{n}{-4} = \frac{1}{\sqrt{53}}$

$\Rightarrow \quad l = \frac{1}{\sqrt{53}}, m = \frac{6}{\sqrt{53}}, n = -\frac{4}{\sqrt{53}}$

Also, $6\left(\frac{1}{\sqrt{53}}\right) + 5\left(\frac{6}{\sqrt{53}}\right) + 9\left(-\frac{4}{\sqrt{53}}\right) = 0$

i.e. line of intersection of perpendicular to normal of third plane. Hence, three planes have a line in common.

90. B

Sol. Lines are x = az + b, y = cz + d

$\Rightarrow \quad \frac{x-b}{a} = \frac{y-d}{c} = \frac{z}{1}$ … (i)

and $x = a_1z + b_1, y = c_1z + d_1$ … (ii)

$\frac{x-b_1}{a_1} = \frac{y-d_1}{c_1} = \frac{z}{1}$

$\therefore$ Lines (i) and (ii) are perpendicular, then

$aa_1 + cc_1 + 1 = 0$